http://bit.ly/digit_HDD
Find the performance table of the six powered drives form this test

Want to make a bootable USB?
To make a Windows 7 bootable USB drive use the popular utility Windows 7 USB/DVD Download Tool available on http://bit.ly/digit_hdd01

you will not bother with that if you can carry 1.5TB of data in your pocket.

Western Digital, the only competitor Seagate has in the internal HDD market, had the maximum capacity drive amongst all the participants with its My Book Essential 3 TB drive. It requires additional power supply, but you can easily access all 3 TB of data unlike the use of a host bus adaptor in its internal drive avataar. The drive has good ventilation across the top and rear edge which helps keep the drive cool under operation. The other two drives from WD were the standard My Passport Essential drives with capacities such as 500GB and 1TB. Both the drives look identical. WD SmartWare utility is bundled in each of these drives which allows you to password protect your data and back up your data. On installation of the SmartWare utility, it will scan your system and segregate your data into System, Pictures, Other, Music, Movies, and Documents in a graphical format. You can can then either backup entire categories or pick and choose folders the next time you connect your drive and Run Backup to instantly backup data.


WD My Book Essential 3 TB

Hitachi Touro Mobile Pro 500GB drive looks exactly like an iPhone4 thanks to its black body and silver lining on the edge with curved edges and flat surfaces. It was the only drive that offered 3GB of cloud storage facility. This in addition to the Hitachi Backup utility allows you to backup your data locally as well as on the cloud. Data on the cloud can be accessed from anywhere as long as you have a net connection. You have an upgrade option to 250 GB at $49 (approx ₹2,000). The Touro houses a fast 7200 RPM drive.

Hitachi Touro Mobile Pro 500GB

Iomega eGo 500GB portable drive has a smart x-shaped rubber covering which protects it from drops or pressure. Buffalo Mini Station 500GB and Apacer AC230 were simple looking drives with glowing lights. Buffalo Mini Station comes with a variety of utilities such as backup utility, secure lock manager, RAMDISK utility and TurboPC with Turbo Copy for speedy transfer.

ADATA sent us two USB 3.0 drives, one portable and one requiring an external power supply. ADATA Classic CH11 is the 1TB portable drive coming in an all white body. It does not come with any pre-installed softwares. The Nobility NH03 is also a 1TB drive which comes with a power supply and more than makes up for the plain jane looks of the Classic CH11. It has a piano black glossy finish and is rectangular in shape with concave edges. There are enough grooves around the edges to keep the drive cool. On the front portion it has a button for backup, which is a nice touch as opposed to going through a wizard to back up your data.

HOW WE TESTED

We divided the external drives into four categories. For Portable Drives we had a) Drives less than or equal to 750 GB and b) Drives greater than 750 GB. Drives that required an external power supply were clubbed together. Finally Flash drives were divided into a) Drives with capacity under 32 GB and b) Drives with a capacity of 32 GB and up. We did not demarcate between USB 2.0 drives and USB 3.0 drives as the cost per GB is not significantly different in either cases and USB 3.0 drives are backward compatible.

The test bed was made up of following components:

Processor: Intel Core i7-2600K @ 3.4GHz

Memory: 2 x 2GB Corsair Dominator DDR3 RAM @ 1333MHz and timing of 9-9-9-24

Motherboard: Intel DP67BG

OS: Windows 7 Ultimate (64-bit)

OS Drive: Intel SSDSA2SH080GIGN (80 GB SSD)

RAID 0 combination: 2 x 300GB WD Velociraptor (SATA 2, 10000 RPM)

Graphics Card: AMD Radeon HD 6850

Power Supply: Corsair HX620W

We installed a clean copy of Windows 7 Ultimate on the Intel 80 GB SSD and installed the synthetic benchmarks such as HD Tune Pro 4.0 and SiSoftware Sandra 2011 on the SSD. Two partitions were created on the WD Velociraptor RAID 0 array and we used it for real life data transfer tests.

We used HD Tune Pro 4.0.1 for testing the Average Read / Write and Read Burst speeds as well as SiSoft Sandra 2011. Both these benchmarks require unformatted drives to perform write testing.

For our Real World tests we used a 4 GB RAR File and a 4GB Assorted file, to simulate a sequential and assorted file transfer scenario. These files were copied to and from the RAID 0 array to the external drives under test. We also performed file transfer tests within the two partitions of the external drives. Finally, we used Adobe Photoshop CS2 to test the amount of time taken to load a 1GB file. Both the image and the scratch disk were kept on the test drives on the same partition, effectively stimulating file access. We did not perform intra-drive file transfer and Photoshop file access on Flash drives.

Performance

We had around four drives crossing the 100 MB/s barrier as far as sequential read and write was concerned. ADATA Nobility NH03 gave very impressive read and write speeds of over 120 MB/s. Seagate GoFlex Slim was another drive that left us impressed with its performance. When it came to Assorted file read, only WD My Book and Nobility NH03 crossed the 100 MB/s. Both these drives are powered, whereas GoFlex Slim and Hitachi Touro were the only portable drives to reach within a breathing distance of 100 MB/s in assorted read tests.